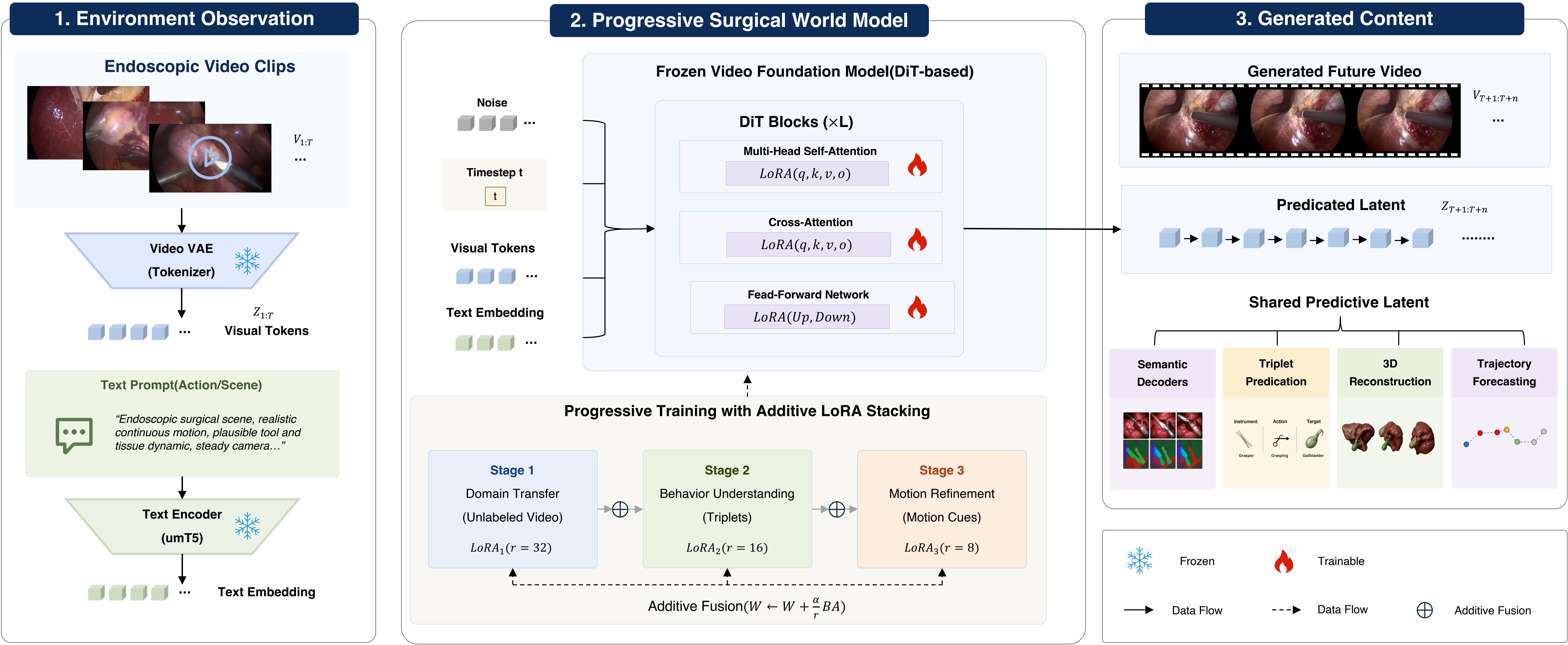

1. Environment Observation
Endoscopic Video Clips
$V_{1:T}$
...
Video VAE
(Tokenizer)
$Z_{1:T}$
Visual Tokens
Text Prompt(Action/Scene)
"Endoscopic surgical scene, realistic continuous motion, plausible tool and tissue dynamic, steady camera..."
Text Encoder
(umT5)
Text Embedding
2. Progressive Surgical World Model
Noise
Timestep t
t
Visual Tokens
Text Embedding
Frozen Video Foundation Model(DiT-based)
DiT Blocks (×L)
Multi-Head Self-Attention
$LoRA(q, k, v, o)$
Cross-Attention
$LoRA(q, k, v, o)$
Fead-Forward Network
$LoRA(Up, Down)$
Progressive Training with Additive LoRA Stacking
Stage 1
Domain Transfer
(Unlabeled Video)
$LoRA_1(r = 32)$
Stage 2
Behavior Understanding
(Triplets)
$LoRA_2(r = 16)$
Stage 3
Motion Refinement
(Motion Cues)
$LoRA_3(r = 8)$
Additive Fusion($W \leftarrow W + \frac{\alpha}{r} BA$)
3. Generated Content
Generated Future Video
$V_{T+1:T+n}$
...
Predicated Latent
$Z_{T+1:T+n}$
....
Shared Predictive Latent
Semantic Decoders
Triplet Predication
Instrument
Action
Target
Grasper
Grasping
Gallbladder
3D Reconstruction
Trajectory Forecasting
Frozen
Trainable
Data Flow
Data Flow
Additive Fusion